

Nolan Koblischke

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Research interests: Evaluation and benchmarking of AI research agents. Building discovery systems over large-scale scientific datasets. Foundation models for scientific data.

EDUCATION

University of Toronto

Ph.D. in Astrophysics

ON, Canada

Sep 2023 – 2028

Thesis: *Foundation Models for Search and Discovery in Survey Astronomy*

University of British Columbia

B.Sc. in Physics with Honours, Minor in Computer Science. GPA: 3.99/4.0

BC, Canada

2019 – 2023

Thesis: *Tip of the Red Giant Branch Calibration using Long-period Variable Stars*

RESEARCH EXPERIENCE

Polymathic AI at NYU and Simons Foundation

New York City, USA

Junior Research Scientist

Summer 2025

Built and deployed a semantic search engine for 100M+ galaxy images using LLM-generated captions and contrastive learning. Led to the discovery of 36 galaxies with stellar streams useful for studying dark matter.

University of Toronto

Toronto, ON, Canada

Graduate Researcher

2023 – Present

Developed a simulation-based benchmark to test agents on experiment planning and discovery in physics.

Currently developing methods for using language models for search and discovery in astrophysics.

École Polytechnique Fédérale de Lausanne (EPFL)

Geneva, Switzerland

Research Intern, Astrophysics

Summer 2022 & 2023

PUBLICATIONS

1. **Nolan Koblischke**, Liam Parker, Francois Lanusse, Irina Espejo Morales, Jo Bovy, Shirley Ho, “Semantic search for 100M+ galaxy images using AI-generated captions,” *NeurIPS 2025 AI4Science Workshop* (Spotlight). To be submitted to Astrophysical Journal. arXiv:2512.11982 [astro-ph.IM] aion-search.github.io
2. **Nolan Koblischke**, Hyunseok Jang, Kristen Menou & Mohamad Ali-Dib, “Gravity-Bench-v1: A Benchmark on Gravitational Physics Discovery for Agents,” *ICML 2025* (Poster). arXiv:2501.18411 [cs.AI] gravitybench.github.io
3. **Nolan Koblischke** & Jo Bovy, “SpectraFM: Tuning into Stellar Foundation Models,” *NeurIPS 2024 Workshop: Foundation Models for Science*. arXiv:2411.04750 [astro-ph.IM]
4. **Nolan Koblischke** & Richard I. Anderson, “Calibrating and Standardizing the Tip of the Red Giant Branch in the Small Magellanic Cloud Using Small-amplitude Red Giants,” *Astrophys. J.* 974, 181 (2024). arXiv:2406.19375 [astro-ph.SR]
5. Richard I. Anderson, **Nolan Koblischke** & Laurent Eyer, “Small-amplitude Red Giants Elucidate the Nature of the Tip of the Red Giant Branch as a Standard Candle,” *Astrophys. J. Lett.* 963, L43 (2024). arXiv:2303.04790 [astro-ph.CO]

SUBMITTED / UNDER REVIEW

1. C. Ye, S. Yuan, S. Cooray, S. Dillmann, I. L. V. Roque, D. Baron, P. Frank, S. Martin-Alvarez, **Nolan Koblischke**, F. J. Qu, D. Yang, R. Wechsler, & I. Ciucă, “ReplicationBench: Can AI Agents Replicate Astrophysics Research Papers?” (submitted). arXiv:2510.24591 [cs.CL]

ACADEMIC SERVICE

Area Chair, NeurIPS Machine Learning and the Physical Sciences Workshop, 2025.

Reviewer, Nature Astronomy, 2025.

Reviewer, NeurIPS AI4Science Workshop, 2025.

Reviewer, The Astrophysical Journal, 2025.

SELECTED TALKS & PRESENTATIONS

- “Semantic search for 100M+ galaxy images using AI-generated captions” — Talk, Harvard-Smithsonian, Center for Astrophysics, 2026. [Link](#)
- Podcast Guest, “Teaching ChatGPT to Do Real Science” — Universe Today, 2025. [Link](#)
- Panelist, “AI & Physics: Learning from Each Other” - Vector Institute Foundation Models for Science, Toronto, 2025.
- “Gravity Bench: A Benchmark for an AI Astronomer” — Talk, Workshop: NLP for Space Science, ESA/ESAC Madrid, 2024. [Link](#)
- “SpectraFM: Tuning into Stellar Foundation Models” — Talk, Workshop: AstroAI, Harvard-Smithsonian CfA, 2024. [Link](#)
- “ChatGaia: Talk to the Gaia Archive in Natural Language” — Talk, Conference: Debating the Potential of Machine Learning in Astronomical Surveys, CCA/Flatiron Institute, 2023. [Link](#)

SELECTED AWARDS & HONORS

OpenAI’s ChatGPT Futures, Class of 2026 (awarded to 26 individuals and teams)	2026
NSERC CGRS D (\$120,000 over 3 years)	2026 – 2028
Data Sciences Institute Doctoral Student Fellowship (\$75,000 over 3 years)	2025 – 2028
FAST Doctoral Award (\$76,000 over 4 years)	2023 – 2027
NSERC CGS M (\$27,000)	2024 – 2025
EPFL Scholarship of Excellence (\$13,000)	2022
UBC Presidential Scholars Award (\$30,000)	2019 – 2023

TECHNICAL SKILLS

- ML / AI:** PyTorch, pre-training scientific foundation models, agent development and evaluation, language model fine-tuning (SFT, RL), vision-language models, contrastive learning.
- Engineering:** Python, Git, HPC clusters (SLURM), LLM APIs, large-scale data pipelines.
- Research:** Astrophysicist by training. Experienced in leading experimental design, analysis, and academic writing.

SELECTED PROJECTS & TOOLS

- LLM Tools for Astronomers
- Developed LLM tools to accelerate access to astronomical data and research.
 - ChatGaia (2023): Natural-language interface to Gaia’s 2 billion star database (70k+ viewers).
 - ChatADS (2023): Literature search assistant for astronomy. (1k+ users)
 - AstroCoder (2025): Code generation tool for astronomical data analysis workflows.